Quartz crystals start to form at temperatures of about **1,700°F (3,090°C)**.

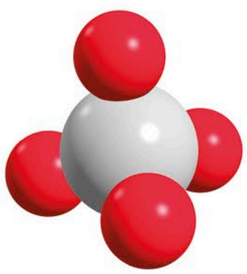
Crystals

A crystal is a solid material, made of atoms set in a repeating 3-D pattern. Crystals form from minerals when molten magma cools to become solid rock. Crystals of some substances, such as salt, sugar, and ice, are formed through evaporation or freezing.

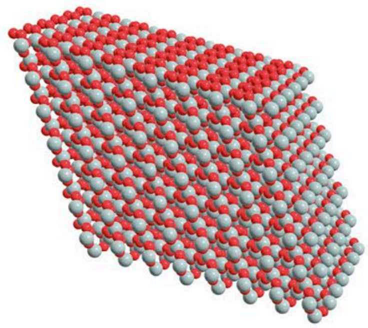
The shapes and colors of mineral crystals depend on the elements from which they are made and the conditions (the temperature and pressure) under which they formed. The speed at which the magma cools decides the size of the crystals. Crystals can change under extreme pressure in the rock cycle (see p.22), when one rock type changes into another.

Crystal structures

Crystals have highly ordered structures. This is because the atoms or molecules in a crystal are arranged in a 3-D pattern that repeats itself exactly over and over again. Most metals have a crystalline structure, too.



Quartz tetrahedron
The molecule that makes up quartz is in the shape of a tetrahedron, made of four oxygen atoms and one silicon atom.



Quartz crystal
A quartz crystal consists of a lattice of tetrahedrons, repeated in all directions.

One mineral, two gem crystals

Crystals of the mineral corundum come in many colors, thanks to different impurities in the crystal structure. Often cut and polished to be used as gems, the best known are sapphire (usually blue) and ruby (red).

BLUE CORUNDUM: SAPPHIRE

RED CORUNDUM: RUBY

CUT RUBY CRYSTAL SET IN A RING

Quartz crystals

The crystal quartz is one of the most common minerals in Earth's crust. It comes in many different forms and colors—but they all share the same formula: silicon dioxide, or SiO_2. Some of the best known include rock crystal (transparent), rose quartz (pink), tiger-eye (yellow-brown), citrine (yellow), and amethyst (purple). Their beauty makes them popular for jewelry, whether in natural form, tumbled, or cut and polished.

Amethyst geode
A geode is formed when gas bubbles are trapped in cooling lava. The crystals lining the walls of the geode grow when hot substances containing silicon and oxygen, as well as traces of iron, seep into the cavities left by the bubbles.

The purple color of amethyst comes from iron impurities in the crystal structure.

The outer shell of the geode is normally a volcanic, igneous rock such as basalt.

Crystal systems

The shape of a crystal is determined by how its atoms are arranged. This decides the number of flat sides, sharp edges, and corners of a crystal. Crystals are sorted into six main groups, known as systems, according to which 3-D pattern they fit.

Cubic
Gold, silver, diamond, the mineral pyrite (above), and sea salt all form cubic crystals.

Tetragonal
Zircon, a silicate mineral, is a typical tetragonal crystal, looking like a square prism.

Hexagonal and trigonal
Apatite is a hexagonal crystal, with six long sides. Trigonal crystals have three sides.

Monoclinic
Orthoclase (above) and gypsum crystals are monoclinic, one of the most common systems.